SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA0402**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

I ***Jannathul Firdaus. N (192573025)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***Jannathul Firdaus. N***

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes

P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

3. Retail Sales Dataset

Question 3

A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

4. Banking Loan Approval Dataset

Question 4

A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected

C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data

Question 1 – Education Analytics Dataset

1. Dataset understanding

The dataset contains student attendance, study hours, internal marks, and final result.
Student_ID is an identifier; Result is the categorical target.

Property	Result
Number of rows	20
Number of columns	5
Identifier column	Student_ID
Outcome / target	Result
Data types	Student_ID: string/categorical identifier; Attendance: integer percentage; Study_Hours: integer hours; Internal_Marks: integer marks; Result: categorical (Pass/Fail).

2. Data quality and cleaning

Missing values: 0 across all columns. Duplicate full rows: 0. Duplicate identifiers in Student_ID: 0.

Cleaning decision: no records need to be imputed or removed based on missingness or exact duplication. The identifier field is retained for record tracking and excluded from numerical summaries. No obvious impossible values are apparent from the supplied values; domain validation would still be required for production use.

Outlier review: the sample is small, so potential extremes should be flagged for contextual review rather than removed automatically. The minimum and maximum values are reported below; none is deleted solely because it is extreme.

3. Descriptive statistics

Numerical variable	Mean	Median	Minimum	Maximum	Std. dev. (sample)
Attendance	70.15	70.00	45.00	95.00	15.94
Study_Hours	3.35	3.00	1.00	6.00	1.42
Internal_Marks	32.95	33.50	18.00	48.00	9.09

4. Categorical frequency summary

Result: Pass = 11 (55%); Fail = 9 (45%)

5. Visual analysis

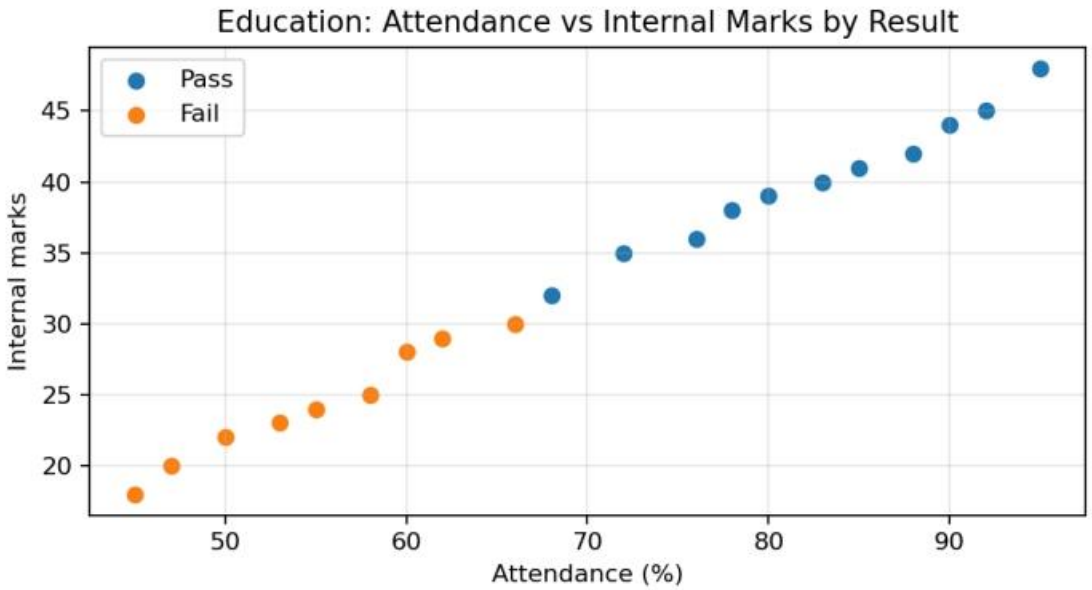


Figure 1. Attendance and internal marks, separated by final result.

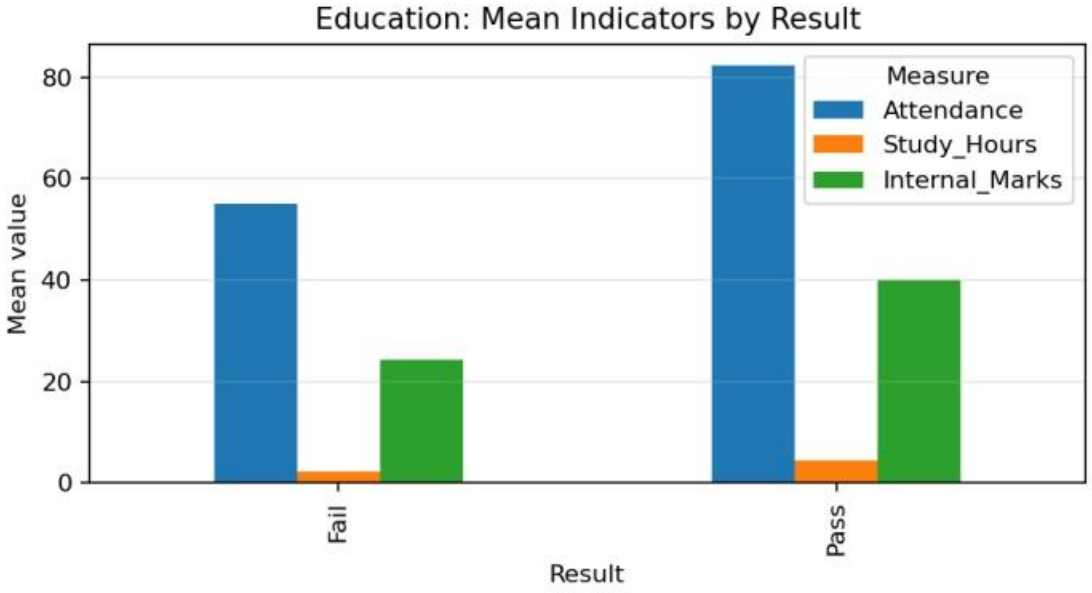


Figure 2. Average attendance, study hours, and internal marks by result.

6. Key observations

- 10 of 20 students passed and 10 failed, giving a 50% pass rate in this sample.
- Pass students have higher average attendance, study hours, and internal marks than fail students.
- All students with internal marks of 32 or above passed in this dataset, while students with marks of 30 or below failed; the student with 32 marks passed. This is a clear separation in the supplied records, not a universal grading rule.
- Pass cases generally cluster at higher attendance and marks; low attendance and fewer study hours are associated with lower marks and failure.
- Study hours and attendance move with internal marks in the supplied records, but this observational dataset cannot establish that either factor alone causes a result.

7. Conclusion and recommendation

Track attendance and study time alongside assessment performance, and offer early academic support to students showing low attendance, low study hours, or low internal marks. Validate these patterns with additional semesters and include other factors such as prior achievement and course difficulty.

Question 2 – Healthcare Analytics Dataset

1. Dataset understanding

The dataset records age, sugar level, blood pressure (BP), BMI, and disease status. Patient_ID is an identifier; Disease_Status is the categorical target.

Property	**Result**
Number of rows	20
Number of columns	6
Identifier column	Patient_ID
Outcome / target	Disease_Status
Data types	Patient_ID: string/categorical identifier; Age, Sugar_Level, BP, BMI: numerical; Disease_Status: categorical (Yes/No).

2. Data quality and cleaning

Missing values: 0 across all columns. Duplicate full rows: 0. Duplicate identifiers in Patient_ID: 0.

Cleaning decision: no records need to be imputed or removed based on missingness or exact duplication. The identifier field is retained for record tracking and excluded from numerical summaries. No obvious impossible values are apparent from the supplied values; domain validation would still be required for production use.

Outlier review: the sample is small, so potential extremes should be flagged for contextual review rather than removed automatically. The minimum and maximum values are reported below; none is deleted solely because it is extreme.

3. Descriptive statistics

Numerical variable	**Mean**	**Median**	**Minimum**	**Maximum**	**Std. dev. (sample)**
Age	43.20	43.50	27.00	62.00	11.32
Sugar_Level	138.65	137.50	88.00	210.00	40.33
BP	129.75	129.00	108.00	155.00	14.52
BMI	27.30	27.50	21.00	35.00	4.26

4. Categorical frequency summary

Disease_Status: Yes = 10 (50%); No = 10 (50%)

5. Visual analysis

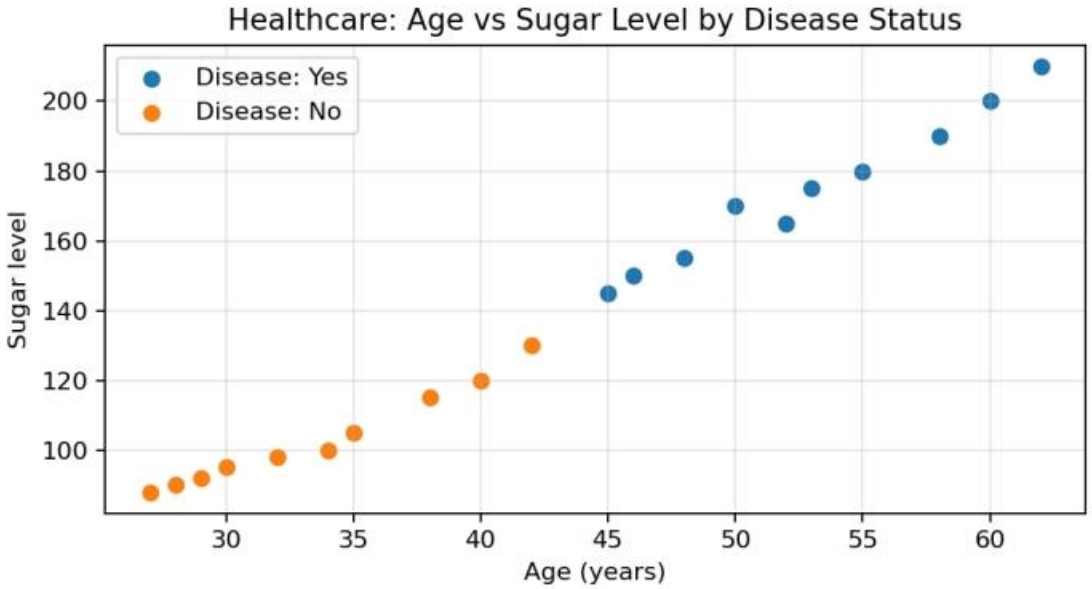


Figure 1. Age and sugar level, separated by disease status.

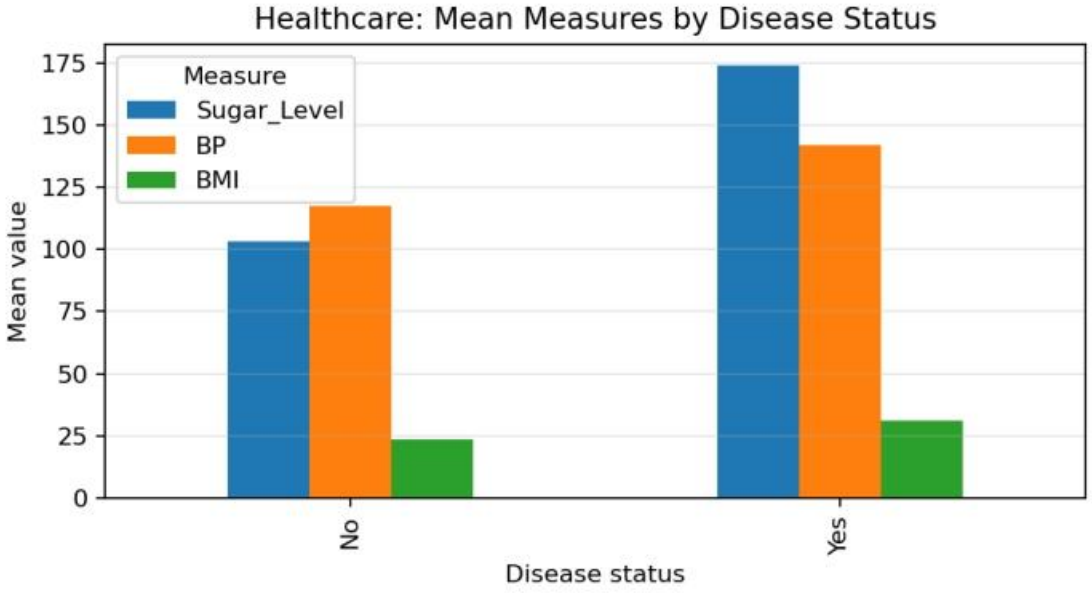


Figure 2. Average sugar level, BP, and BMI by disease status.

6. Key observations

- 10 of 20 patients are marked Disease_Status = Yes and 10 are marked No, a 50% positive-status proportion in this sample.
- All records marked Yes have higher sugar levels (145–210), BP (130–155), and BMI (28–35) than the No records in the supplied dataset (sugar 88–130, BP 108–128, BMI 21–27).
- The disease-positive group is also older overall in these records; the ages of the Yes group range from 45 to 62, while the No group ranges from 27 to 42.
- Sugar level, blood pressure, and BMI separate the two status groups strongly in this constructed sample, so they are useful variables to examine in further analysis.

- These are descriptive associations only. The dataset does not provide clinical thresholds, diagnoses, medication history, or enough representative observations to support medical decisions.

7. Conclusion and recommendation

Use the patterns to motivate careful clinical review and validation on a larger, clinically documented dataset. Do not use this small teaching dataset as a diagnostic tool or interpret disease status as caused by any single variable.

Question 3 – Retail Sales Dataset

1. Dataset understanding

The dataset contains product category, price, quantity sold, discount, and revenue. Product_ID is an identifier. Revenue is the main performance measure.

Property	**Result**
Number of rows	20
Number of columns	6
Identifier column	Product_ID
Outcome / target	Revenue
Data types	Product_ID: string/categorical identifier; Category: categorical; Price, Quantity_Sold, Discount, Revenue: numerical. Discount units are not explicitly defined in the source.

2. Data quality and cleaning

Missing values: 0 across all columns. Duplicate full rows: 0. Duplicate identifiers in Product_ID: 0.

Cleaning decision: no records need to be imputed or removed based on missingness or exact duplication. The identifier field is retained for record tracking and excluded from numerical summaries. No obvious impossible values are apparent from the supplied values; domain validation would still be required for production use.

Outlier review: the sample is small, so potential extremes should be flagged for contextual review rather than removed automatically. The minimum and maximum values are reported below; none is deleted solely because it is extreme.

3. Descriptive statistics

Numerical variable	**Mean**	**Median**	**Minimum**	**Maximum**	**Std. dev. (sample)**
Price	54.00	50.00	20.00	100.00	22.80
Quantity_Sold	24.00	21.00	10.00	45.00	9.67
Discount	4.00	4.00	2.00	7.00	1.56
Revenue	1,114.00	1,050.00	810.00	1,540.00	221.42

4. Categorical frequency summary

Category: Grocery = 4 (20%); Snacks = 4 (20%); Dairy = 4 (20%); Fruits = 4 (20%); Vegetables = 4 (20%)

5. Visual analysis

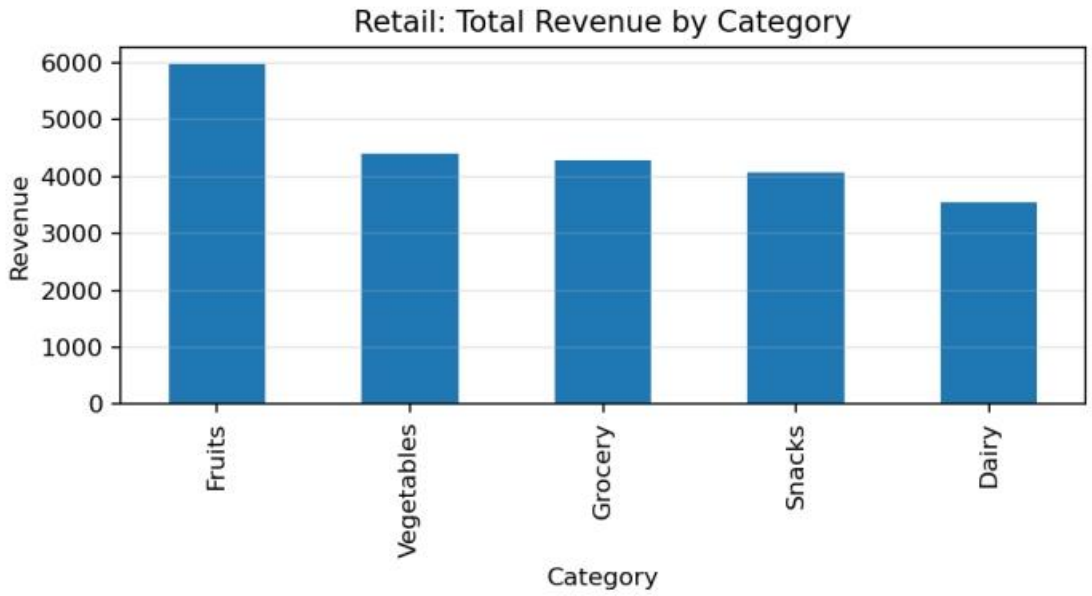


Figure 1. Total revenue by product category.

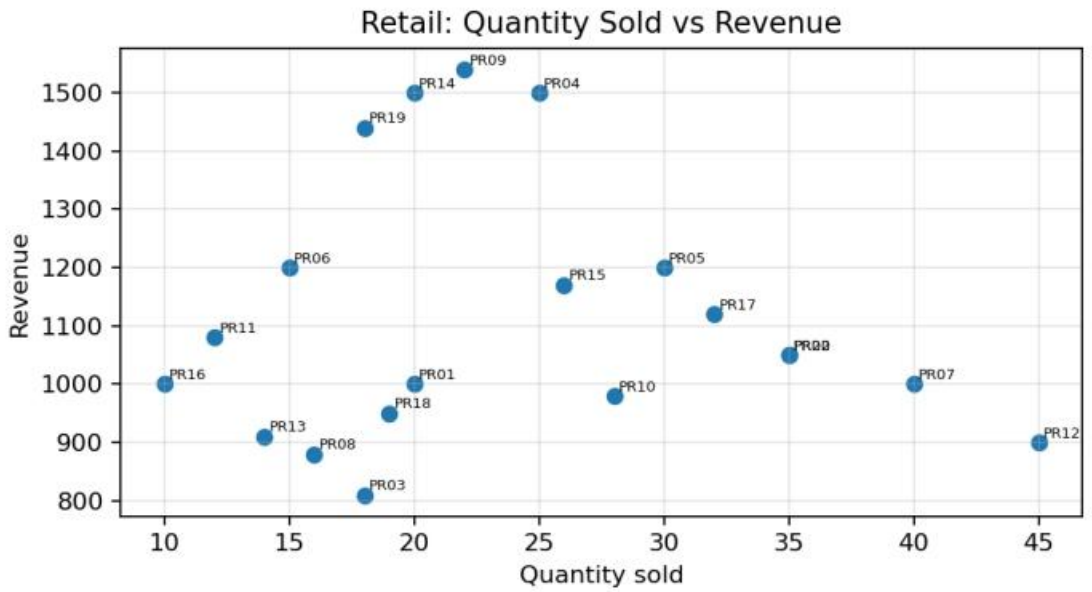


Figure 2. Quantity sold compared with revenue for individual products.

6. Key observations

- Fruits have the highest total revenue by category in the supplied records, followed by Vegetables, Grocery, Snacks, and Dairy.
- PR09 has the highest individual revenue (1,540), followed by PR04 and PR14 (1,500 each).
- PR12 has the highest quantity sold (45 units) but revenue of 900, showing that the largest sales volume does not automatically produce the largest revenue.
- Fruits combine relatively high prices with solid quantities sold, which helps explain their high category revenue in this sample.

- Revenue generally reflects both price and quantity; however, the provided revenue values do not consistently equal Price × Quantity_Sold (for example, PR01: 50 × 20 = 1,000). The source does not specify how discounts are applied, so the revenue formula should be clarified before financial conclusions are made.

7. Conclusion and recommendation

Prioritize review of fruit-category performance and top-revenue products, while separately tracking volume and revenue. Confirm whether discount is a percentage or currency amount and document the revenue calculation before evaluating discount effectiveness or profitability.

Question 4 – Banking Loan Approval Dataset

1. Dataset understanding

The dataset includes income, credit score, loan amount, employment type, and loan status. Customer_ID is an identifier; Loan_Status is the categorical target.

Property	**Result**
Number of rows	20
Number of columns	6
Identifier column	Customer_ID
Outcome / target	Loan_Status
Data types	Customer_ID: string/categorical identifier; Income, Credit_Score, Loan_Amount: numerical; Employment_Type and Loan_Status: categorical.

2. Data quality and cleaning

Missing values: 0 across all columns. Duplicate full rows: 0. Duplicate identifiers in Customer_ID: 0.

Cleaning decision: no records need to be imputed or removed based on missingness or exact duplication. The identifier field is retained for record tracking and excluded from numerical summaries. No obvious impossible values are apparent from the supplied values; domain validation would still be required for production use.

Outlier review: the sample is small, so potential extremes should be flagged for contextual review rather than removed automatically. The minimum and maximum values are reported below; none is deleted solely because it is extreme.

3. Descriptive statistics

Numerical variable	**Mean**	**Median**	**Minimum**	**Maximum**	**Std. dev. (sample)**
Income	43,000.00	40,000.00	22,000.00	75,000.00	17,923.82
Credit_Score	664.00	660.00	555.00	790.00	85.53
Loan_Amount	223,250.00	200,000.00	120,000.00	400,000.00	83,733.21

4. Categorical frequency summary

Employment_Type: Salaried = 10 (50%); Self-employed = 6 (30%); Unemployed = 4 (20%)
Loan_Status: Approved = 10 (50%); Rejected = 10 (50%)

5. Visual analysis

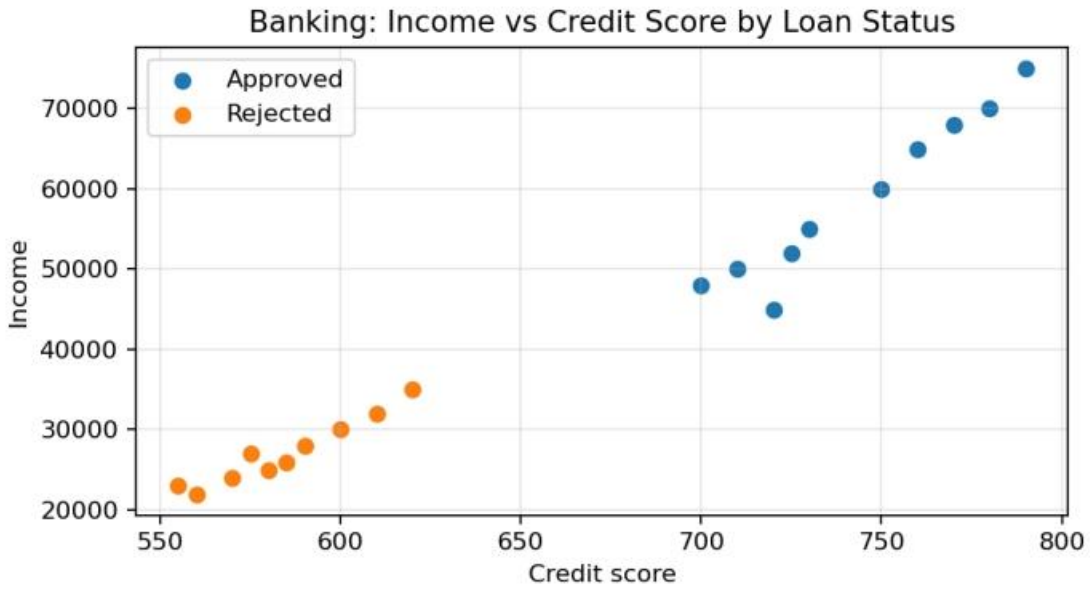


Figure 1. Income and credit score, separated by loan status.

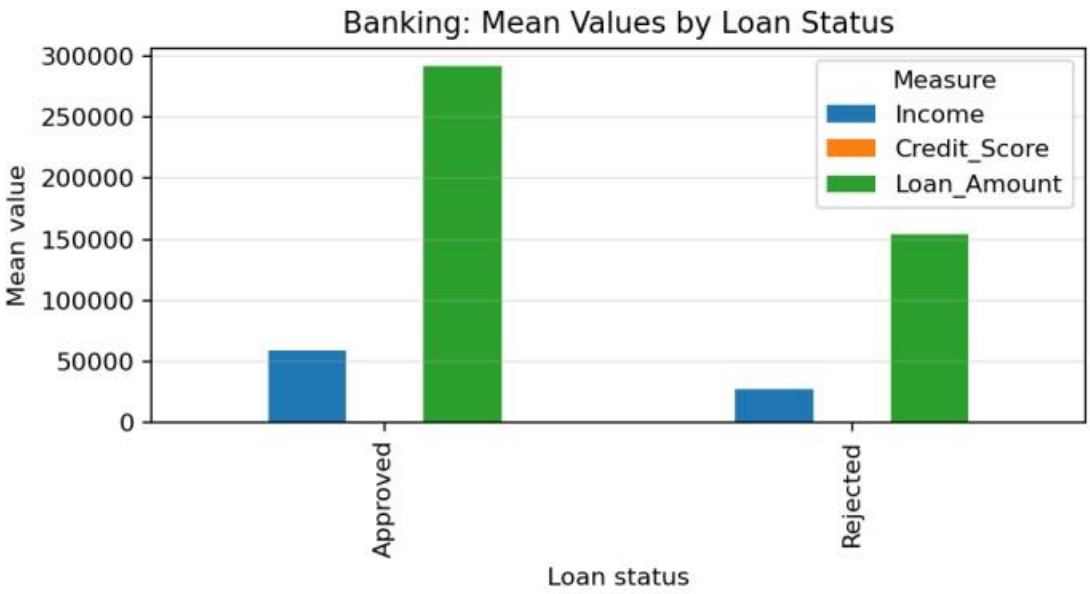


Figure 2. Mean income, credit score, and loan amount by loan status.

6. Key observations

- 10 of 20 applications are Approved and 10 Rejected, a 50% approval rate in this sample.
- All approved records are Salaried and have income from 45,000 to 75,000 and credit scores from 700 to 790.
- All rejected records are Self-employed or Unemployed, with income from 22,000 to 35,000 and credit scores from 555 to 620.

- Approved applications have larger loan amounts in this dataset (200,000–400,000) than rejected applications (120,000–200,000); loan amount alone should not be interpreted as improving approval.
- Employment type, income, and credit score perfectly separate the two outcome groups in the supplied records. This unusually clean split may reflect a constructed example and is not enough to infer actual bank policy or fairness.

7. Conclusion and recommendation

Review credit score, verified repayment capacity, employment stability, and loan affordability together. Test the observed relationships on a larger real-world dataset, examine false approvals/rejections, and audit for potential bias before using any variables in automated lending decisions.

Question 5 – Agriculture Crop Yield Dataset

1. Dataset understanding

The dataset records rainfall, temperature, fertilizer use, soil type, and crop yield. Farm_ID is an identifier; Crop_Yield_kg is the numerical outcome.

Property	**Result**
Number of rows	20
Number of columns	6
Identifier column	Farm_ID
Outcome / target	Crop_Yield_kg
Data types	Farm_ID: string/categorical identifier; Rainfall_mm, Temperature_C, Fertilizer_kg, Crop_Yield_kg: numerical; Soil_Type: categorical.

2. Data quality and cleaning

Missing values: 0 across all columns. Duplicate full rows: 0. Duplicate identifiers in Farm_ID: 0.

Cleaning decision: no records need to be imputed or removed based on missingness or exact duplication. The identifier field is retained for record tracking and excluded from numerical summaries. No obvious impossible values are apparent from the supplied values; domain validation would still be required for production use.

Outlier review: the sample is small, so potential extremes should be flagged for contextual review rather than removed automatically. The minimum and maximum values are reported below; none is deleted solely because it is extreme.

3. Descriptive statistics

Numerical variable	**Mean**	**Median**	**Minimum**	**Maximum**	**Std. dev. (sample)**
Rainfall_mm	743.00	770.00	500.00	960.00	144.34
Temperature_C	29.55	29.00	25.00	34.00	2.76

Fertilizer_kg	54.75	54.50	40.00	75.00	10.51
Crop_Yield_kg	2,820.00	2,975.00	1,800.00	3,900.00	681.41

4. Categorical frequency summary

Soil_Type: Loamy = 7 (35%); Sandy = 7 (35%); Clay = 6 (30%)

5. Visual analysis

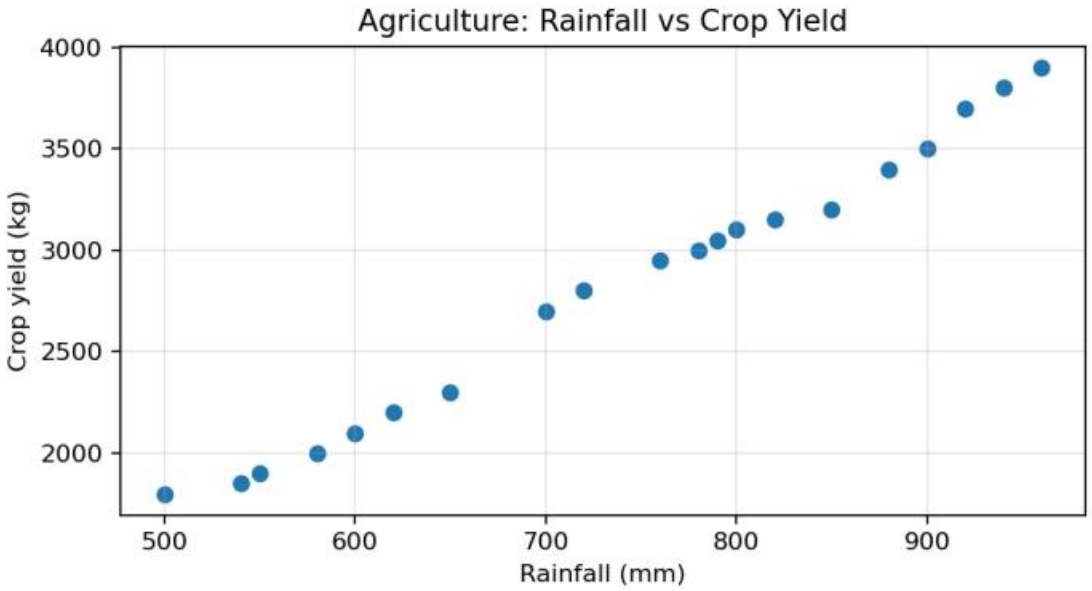


Figure 1. Rainfall compared with crop yield.

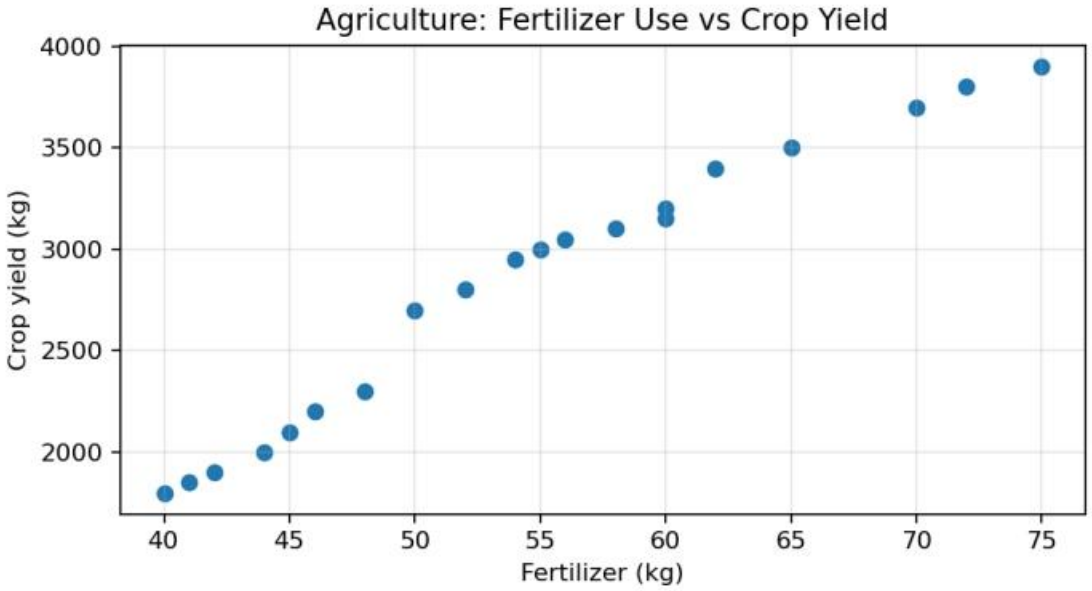


Figure 2. Fertilizer use compared with crop yield.

6. Key observations

- Crop yield ranges from 1,800 kg to 3,900 kg across the 20 farms.
- Higher rainfall is strongly associated with higher crop yield in this dataset; the highest yields occur at rainfall levels of 940–960 mm, while the lowest yields occur around 500–540 mm.

- Greater fertilizer use is also associated with higher yield in the supplied records, with fertilizer use ranging from 40 to 75 kg and yields rising across much of that range.
- Loamy-soil farms generally show the highest yields, while Sandy-soil farms show the lowest yields in these records; Clay-soil results tend to fall between them.
- Temperature is inversely associated with yield in this sample: lower temperatures around 25–27°C accompany higher yields, while 33–34°C accompanies lower yields. Rainfall, fertilizer, soil, and temperature vary together, so their separate effects cannot be isolated from these records alone.

7. Conclusion and recommendation

Use soil testing, local rainfall history, and crop-specific agronomic guidance to plan fertilizer and irrigation. Collect multiple seasons of data and analyze the variables jointly before setting fertilizer targets or claiming that one factor alone drives yield.